

# Day 15: DAX Time Intelligence Mastery

## 📌 Definition to Remember

**Time Intelligence functions** in DAX calculate metrics over specific calendar periods including Year-To-Date (**YTD**), Prior Year comparisons (**YoY Growth**), and moving rolling averages.

## 1. Prerequisites for Time Intelligence

- Dedicated Date Table ( **Dim\_Date** ) with contiguous, unbroken dates.
- Mark as Date Table verified in Power BI Desktop.

## 2. Year-To-Date (YTD) & Month-To-Date (MTD)

```
// Year To Date Sales
Sales YTD = TOTALYTD([Total Sales], Dim_Date[Date])

// Month To Date Sales
Sales MTD = TOTALMTD([Total Sales], Dim_Date[Date])
```

## 3. Prior Year & YoY Growth % Calculations

```
// Previous Year Sales (Same Period Last Year)
Sales PY =
CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR(Dim_Date[Date])
)

// Year-over-Year (YoY) Sales Variance
YoY Sales Growth $ = [Total Sales] - [Sales PY]

// Year-over-Year (YoY) Sales Growth %
YoY Sales Growth % =
DIVIDE([YoY Sales Growth $], [Sales PY], 0)
```

## 4. Moving Rolling Averages (30-Day Rolling Sales)

```
Rolling 30 Days Sales =
CALCULATE(
    [Total Sales],
    DATESINPERIOD(Dim_Date[Date], MAX(Dim_Date[Date]), -30, DAY)
)
```

### ★ Must-Master Skills & Interview Points

1. `SAMEPERIODLASTYEAR()` vs `DATEADD(..., -1, YEAR)` .
2. How `DATESYTD` / `TOTALYTD` automatically handle leap years and fiscal year-ends ( `"3/31"` ).
3. Formatting YoY growth with dynamic up/down color indicators ( ▲ / ▼ ).

### ⚡ Quick Recall

`TOTALYTD / TOTALMTD + SAMEPERIODLASTYEAR (Prior Year) + DATESINPERIOD (Rolling Averages)`